Essays

|!|BCEEB0190000001|!| ★ **Cross topic**

Give an account on how the pancreas is involved in the rise of the blood glucose level after a carbohydrate-rich meal and the related homeostatic control. (11 marks)

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Rise in the blood glucose level after a carbohydrate-rich meal (4)

- The pancreas secretes pancreatic juice to the duodenum through the pancreatic duct. (1)
- Pancreatic juice contains pancreatic amylase which catalyses the breakdown of starch in the chyme to maltose. (1)
- Maltose is then further broken down into glucose in the small intestine. (1)
- A lot of glucose is absorbed into the bloodstream from the small intestine; hence the blood glucose level increases. (1)

Regulation of the blood glucose level (Max. 4)

- A rise in the blood glucose level is detected by the pancreas. (1)
- In response, the pancreas secretes more insulin and less glucagon into the blood. (1)
- Insulin stimulates the liver cells to convert excess glucose to glycogen. The glycogen is then stored in the liver and muscles. (1)
- Insulin also stimulates the uptake of glucose by body cells and subsequent oxidation of glucose by respiration. (1)
- As a result, the blood glucose level falls and returns to normal. (1)

Communication (Max. 3)

(11 marks)

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Skills assessed	Marks
Understanding basic concepts	8
Communication	3

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|!|BCEEC0190000002|!|

State the different components of the negative feedback mechanism in the regulation of blood glucose level and explain how the blood glucose level is regulated in humans.

(11 marks)

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Main components of negative feedback mechanism in the regulation of blood glucose level (Max. 3)

- Receptor: pancreas (1)
- Control centre: pancreas (1)
- Effector: liver cells / body cells (1)

When blood glucose level is higher than normal (Max. 3)

- The pancreas detects a rise in the blood glucose level. (1)
- It secretes more insulin and less glucagon into the blood. (1)
- Insulin stimulates the liver cells to convert excess glucose to glycogen. (1)
- Insulin stimulates the uptake of glucose by body cells and the subsequent oxidation of glucose by respiration. (1)
- As a result, the blood glucose level falls and returns to normal.

When blood glucose level is lower than normal (Max. 2)

- The pancreas detects a fall in the blood glucose level. (1)
- It secretes more glucagon and less insulin into the blood. (1)
- Glucagon stimulates the liver cells to convert glycogen to glucose. (1)
- As a result, the blood glucose level rises and returns to normal.

Communication (Max. 3)

(11 marks)

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Skills assessed	Marks
Understanding basic concepts	8
Communication	3

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